

Course Project Proposal: Evaluating and Improving the Robustness of Ultra-Lightweight Transformers for Extreme Super-Resolution

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Abstract

This document outlines the scoping and planning for a computer vision course project. The project aims to investigate the limitations of state-of-the-art lightweight transformer models in Single Image Super-Resolution (SISR), specifically focusing on the Efficient Attention-Sharing Information Distillation Transformer (ASID). We identify a research gap regarding the model's performance under extreme upscaling factors ($\times 8$) and propose a lightweight pre-upsampling intervention to mitigate performance degradation. All relevant literature, datasets, and project plans are consolidated herein.

1 Paper Discovery and Literature Review

The following table summarizes the core research papers selected during the discovery phase. The primary implementation paper is ASID (AAAI 2025), surrounded by foundational baselines and contemporary state-of-the-art models across IEEE, Elsevier, Springer, ACM, and top-tier conferences (CVPR, ICCV, ECCV).

2 Research Gap Synthesis

Across the shortlisted papers, several concrete, buildable research gaps were identified. The primary gap selected for this course project is highlighted below.

2.1 Selected Gap: Real-world / Blind-degradation generalization of ultra-lightweight SR

- **Description:** The ASID paper validates its model only on bicubic $\times 2$ and $\times 4$ upscaling with 313K parameters. Behavior under extreme upscaling ($\times 8$) and blind Real-ESRGAN-style degradation is untested. DiT-SR explicitly notes that from-scratch SR "still has some way to go" vs. prior-based models.
- **Feasibility:** Very high — 313K parameters, DIV2K dataset (800 images), trains in hours on a single free GPU (Google Colab).
- **Novelty:** Medium — Real-world SR is heavily studied, but for this specific attention-sharing design, it is completely untested.
- **Implementation Effort:** Low-Medium.

#	Title & Authors	Venue, Year	Role	Pa-	Link
1	Efficient Attention-Sharing Information Distillation Transformer for Lightweight SISR (ASID) — Park, Soh, Cho	AAAI, 2025	Main paper		arXiv / Code
2	Enhanced Deep Residual Networks for SISR (EDSR) — Lim et al.	CVPRW, 2017	Baseline		arXiv
3	Very Deep Residual Channel Attention Networks (RCAN) — Zhang et al.	ECCV, 2018	Baseline		arXiv
4	Lightweight SISR with Information Multi-Distillation (IMDN) — Hui et al.	ACM MM, 2019	Baseline		arXiv
5	SwinIR: Image Restoration Using Swin Transformer — Liang et al.	ICCVW, 2021	Transformer Baseline		arXiv
6	Efficient Long-Range Attention Network (ELAN) — Zhang et al.	ECCV, 2022	Peer		arXiv
7	Omni Aggregation Networks for Lightweight SISR (Omni-SR) — Wang et al.	CVPR, 2023	Peer		arXiv
8	Hybrid Attention Transformer (HAT) — Chen et al.	CVPR, 2023	Peer		arXiv
9	ConvFormer-Based Super-Resolution (CFSR) — Wu et al.	IEEE TIP, 2024	Peer		IEEE Xplore
10	Progressive Focused Transformer (PFT) — Long et al.	CVPR, 2025	Peer		CVPR PDF

Table 1: Selected research papers for the Super-Resolution project.

Dataset	Size	Modality	Public / License
ImageNet-1K	1.28M img	RGB	Public, CC-BY-NC
COCO	118K img	RGB	Public, CC-BY
ACDC	100 pts	Cardiac MRI	Public
Left Atrium (LA)	100 3D MRI	MRI	Public, CC-BY
Pancreas-CT (NIH)	82 CT	CT	Public
MVTec-AD	5,354 img	RGB	Public, CC-BY-NC-SA
VisA	10,821 img	RGB	Public
THUMOS'14	~20h video	RGB+flow	Public
EPIC-Kitchens-100	100h video	RGB+audio	Public, CC-BY-NC
LSDIR	86,991 img	RGB	Public
DIV2K	1,000 img	RGB	Public, CC-BY
RealSR	100 img	RGB	Public
Set5 / Set14	5/14 img	RGB	Public
Urban100 / Manga109	100/109 img	RGB	Public
VAW	~12k img	RGB	Public
SemanticKITTI	~23 scans	3D point cloud	Public, CC-BY-NC
nuScenes-lidarseg	1000 scenes	3D point cloud	Public, CC-BY-NC

Table 2: Datasets identified in the literature review.

3 Dataset Shortlist

3.1 A. All Datasets Used in Source Papers

3.2 B. Filtered Subset for Course Project

The following datasets are free, publicly downloadable without special request, and suitable for a single-semester course project on limited compute (single GPU/Colab).

Dataset	Size	License	Why it fits
DIV2K	1,000 img	CC-BY	Tiny SR training set; ASID trains on it directly.
Set5/Set14/Urban100/Manga109	5/14/100/109 img each	Public	SR eval only; trivial compute.
MVTec-AD	5,354 img	CC-BY-NC-SA	Small, free, standard UAD benchmark.
ACDC	100 pts	Public	Cardiac seg; 2D slices fit Colab.

Table 3: Filtered datasets for single-GPU course project.

4 Final Recommendation and Project Pitch

Recommended Combination:

Paper: ASID (AAAI 2025)

Gap: Real-world / Extreme upscaling robustness of ultra-lightweight SR

Dataset: DIV2K (training) + Set5, Set14, Urban100, Manga109 (evaluation)

4.1 Project Pitch

- **Problem:** The ASID paper achieves state-of-the-art lightweight super-resolution but only tests easy upscaling tasks ($\times 2$, $\times 4$). It is unknown how the attention-sharing mechanism

handles extreme upscaling ($\times 8$) where the receptive field is severely taxed.

- **Methodology:** We will train the ASID model from scratch for $\times 8$ upscaling and observe the PSNR drop compared to $\times 4$.
- **Proposed Improvement:** We will implement a simple "Pre-Upsampling" step. Before feeding the tiny low-resolution image into the ASID network, we use standard bilinear interpolation to double its size ($\times 2$), then let ASID do the final $\times 4$ upscaling.
- **Expected Measurable Gain:** We will compare the PSNR/SSIM of "Direct $\times 8$ ASID" vs "Bilinear $\times 2$ + ASID $\times 4$ ". The goal is to recover a measurable slice of the PSNR drop caused by extreme scaling on Urban100 and Manga109.

5 Quick-Access Project Resources

- **ASID Code (Starting Point):** <https://github.com/saturnian77/ASID>
- **SwinIR Evaluation Utilities:** <https://github.com/jingyunliang/swinir>
- **DIV2K Dataset:** <https://data.vision.ee.ethz.ch/cvl/DIV2K/>
- **Standard SR Test Sets:** Bundled in the ASID/SwinIR/Omni-SR repos.